

Electronic Door Lock

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Home burglary incidents are on the rise. Homes without proper security measures in place are particularly more vulnerable.

Test Points

Test point	Details
TP0	0V, GND
TP1	12V
TP2	5V
TP3	High (when S1 through S3 pressed sequentially)
TP4	High for approx. one second (when S1 through S4 pressed sequentially)

But, it is easy to avoid home break-ins by using a simple solution like the electronic door lock presented here. With this circuit installed, the entry door of your house can be unlocked only by entering a pre-decided password, which helps to prevent unauthorised unlocking. The circuit is very simple and can be assembled using readily available components.

Circuit and working

Fig. 1 shows the circuit of electronic door lock. The circuit is built around dual J-K flip-flop 74LS73 (IC3 and IC4), regulators 7812 and 7805 (IC1 and IC2), regulators 7812 and 7805 (IC1 and IC2), 555 timer (IC6) and a few

other components.

The circuit requires 12V for driving motor M1 and 5V for powering rest of the circuit. The mains power supply is stepped down to 15V, 500mA by transformer X1. This stepped-down AC voltage is rectified by bridge rectifier BR1 and filtered by capacitor C1 before it is fed to IC1. IC1 Regulator IC1 provides regulated 12V DC supply, which is further fed to IC2 to get 5V supply. The glowing of LED1 indicates presence of power in the circuit.

The door opens when switches S1 through S4 are sequentially pressed.

The password is recognised using three flip-flops and transistor T2. Flip-flop FF2 of IC3 is operated in toggle mode by making J2=K2=1 (high).

When S1 is pressed, a ground pulse is provided by the keypad, which makes the output of flip-flop (FF2 of IC3) toggle from logic 0 to 1. This pulls up the inputs of flip-flop FF1 of IC4 to high, thus making it also operate in toggle mode.

When switch S2 is pressed, flip-flop FF1 of IC4 toggles from logic 0 to 1, which, in turn, puts FF2 of IC4 in toggle mode.

When switch S3 is pressed, flip-

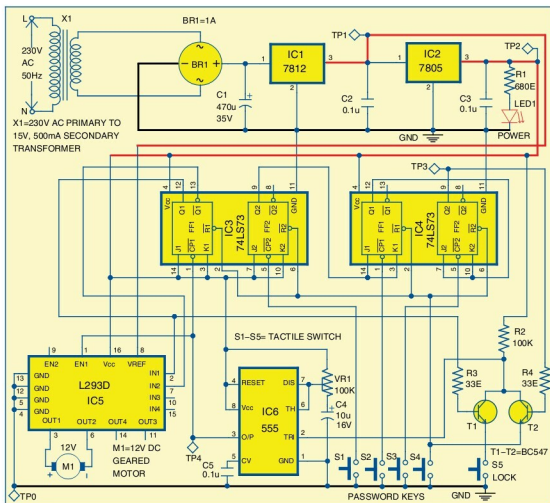


Fig. 1: Circuit of electronic door lock